# Football and Racism: Architecture Write Up

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## 1 Our Overall Approach:

We start our project with the creation of an implicit racism classifier which will be trained and tested on social media comment databases such as football YouTube channels or Reddit football communities. Our main goal with the creation of the classifier is that it can accurately identify and categorize implicit racist discourse within football-related social media content. By fine-tuning advanced Large Language Models such as BERT and LLama2, we will train our system using a rigorously annotated dataset that captures the subtleties of implicit racism. This tool will be at the forefront of our analysis, working in tandem with sentiment analysis, NER, and topic modeling to provide a comprehensive understanding of both overt and subtle forms of racism. Through iterative evaluation and refinement, our classifier aims to be a significant asset in the fight against racism in online football communities. After we have a refined model we will expand our datasets to other social medias to keep building on the foundation of our model.

## 2 Data Collection Details:

In our project, we've adopted a targeted approach to amass a rich and varied dataset centered around football conversations, with a particular emphasis on identifying instances of racism. We harness the YouTube Data API, integrating advanced Python scripting, to systematically capture comments from key football channels. The comments will be sourced from Football channels from the Big Five football associations from England, Germany, Spain, Italy, and France. We will also source comments from top football community YouTube pages such as, Sky Sports Premier League, Tifo Football, HITC Sevens, as well as Euro Football Daily, which will offer a broad spectrum of fan opinions and discussions. For Reddit we will take comment data from the largest communities like r/Soccer, r/PremierLeague, r/Football, etc. The python code will accesses API URL's to make a JSON dataset of YouTube comments. For Reddit, we've chosen a strategic approach, utilizing both the Reddit API and the Python Reddit API Wrapper (PRAW) to mine comments from the most influential football-related subreddits including r/Soccer and r/PremierLeague.

This allows us to tap into the raw pulse of football fan communities globally. Our data collection is further augmented by sophisticated web scraping techniques, where we employ Python's BeautifulSoup to extract insightful articles from leading sports news portals. These articles will provide a more structured and journalistic perspective on the issue of racism in football. In conclusion, our data collection is not just about volume but also about variety and depth. We're not solely gathering data; we are curating a comprehensive dataset that captures the multifaceted nature of football-related discourse, providing a solid foundation for our subsequent analysis and NLP-based investigations.

**Data Processing and Storage Applications:** For database management, we're considering PostgreSQL for its robust data processing capabilities and compliance with SQL standards. We are also considering MongoDB for its efficiency with large volumes of unstructured data. This combination makes sure we have secure, organized, and accessible data storage, vital for our analysis needs.

**Data Annotation:** After accessing the databases through our relational database management system, we will review and annotate data into different sentiment labels (racist, not racist, implicitly racist) for model learning. These individual datasets will then be text preprocessed and feature extracted before being fed into our implicit racism classifier.

# 3 Detailed Explanation of Preprocessing and NLP Analysis Layer:

**Text Preprocessing:**

**Tokenization:** Using libraries like NLTK or spaCy, we break down the text into individual words or tokens. This step is fundamental for understanding the basic structure of the sentences.

**Stemming:** We apply stemming to reduce words to their root form, thereby reducing the complexity of the text data. For instance, "running" becomes "run". NLTK provides efficient tools for this process.

**Lemmatization:** Going a step beyond stemming, lemmatization considers the context and converts the word to its meaningful base form. For example, "better" is converted to "good". spaCy is particularly useful for accurate lemmatization.

**Feature Extraction:**

**Bag-of-Words (BoW):** We convert text documents into a bag-of-words model, which counts the frequency of each word. This simple representation is useful for initial analyses.

**TF-IDF (Term Frequency-Inverse Document Frequency):** This statistical measure evaluates how relevant a word is to a document in a collection of documents. It helps in down-weighting terms that appear frequently across documents.

**Word Embeddings:** We use word embeddings like Word2Vec or GloVe to

capture contextual information within the text. These embeddings provide a dense representation of words based on their contextual meanings.

# 4 Architecture Diagram:

We had an issue with putting the screenshots in the document without them randomly appearing in different sections of the paper, so we have instead provided the link to the full architecture diagram on the Mycourses submission. We believe this issues happened because the multiple screenshots messed with the latex formatting.

# 5 Software entity interactions:

Below are the models we picked that made the most sense to build an implicit racism classifier. We built the model using BERT and LLama2 because BERT is a great model for taking care of the context inside the football related comments and LLama2's analysis can help analyze more nuanced or complex comments. Together this models will be a driving force in capturing the subtleties of implicit racism online.

**NLP Models for Advanced Text Analysis:**

**BERT (Bidirectional Encoder Representations from Transformers):** BERT, a transformer-based model, is extremely effective for understanding the context of a word in a sentence. It will be pivotal in analyzing sentence structures and meanings in our data.

**LLama2:** This model will be used for more nuanced text analysis, especially useful for understanding complex language patterns and nuances in football-related discussions.

**Building the Classifier:** We will train the classifier using the preprocessed data, employing the NLP models above (BERT, LLama2).The training process will involve fine-tuning these models on our annotated dataset, enabling the classifier to discern subtle language cues and contextual meanings associated with implicit racism.The classifier's performance will be continuously evaluated and improved upon, ensuring high accuracy and reliability in detecting implicit racism in football discourse.

**Classification and Sentiment Analysis Summary:** In this phase, we apply machine learning models for text classification and sentiment analysis. For classification, we use models like Support Vector Machines and Random Forest to categorize text into 'racist', 'not racist', and 'implicitly racist'. These models are chosen for their effectiveness in handling complex linguistic patterns in large datasets. For sentiment analysis, techniques such as Naive Bayes and LSTM networks, potentially including pre-trained models like BERT, will be utilized to assess the emotional tone of the content. This dual approach of classification and sentiment analysis is central to accurately identifying and understanding the nuances of racism in football-related discourse.

**Visualization and Reporting Layer Overview:** In the Visualization and Reporting Layer, we leverage Tableau, a powerful data visualization tool, to effectively present the insights gleaned from our analysis. Tableau's robust features enable us to create interactive and visually appealing dashboards and reports that succinctly convey complex data patterns. This includes the distribution of sentiment across various sources, the prevalence of specific language patterns indicative of implicit racism, and comparative analyses across different football associations and platforms. By utilizing Tableau, we ensure that our findings are not only insightful but also accessible and engaging to a wide range of stakeholders, facilitating informed decision-making and awareness around the issue of racism in football.

# 6 Project Navigation and Error Management:

As a default, any flagged errors will be dealt with accordingly to the error message, but in order to catch errors that are unflagged, we utilize our error management process. Our process includes frequent printing checks of data outputs in the early stages of preprocessing and feature extraction in order to make sure the output aligns with our expected outlook. Moving on to the model pipeline we will regularly make sure that any data transformations, sentiment analysis outlook, or model accuracy results live up to our reviewed standards. If any of these outputs fail to give the level of results that is expected of such a sensitive subject, we will rerun the code with added implementations to get as close to our expected results as possible. If we are unable to push our results closer to what we expect, we will communicate with our professional advisor Dr. KhudaBukhsh. Any disagreement between teammates Modal and Meenakshi will be resolved through a mediation process hosted by the professor if need be. Most if not all disagreements should inherently be resolved within the group, using open, honest and respectful discussion to tackle any issues that may arise.

# 7 Conclusion:

In conclusion, our project stands at the intersection of data science and social advocacy, wielding the power of NLP and machine learning to confront racism in football. By meticulously collecting and processing data, developing an acute racism classifier, and transforming findings into visual narratives, we endeavor to shed light on the often-overlooked nuances of discriminatory language. Our work is not merely an academic exercise; it's a step towards a more inclusive and respectful football culture. As we move forward, we remain committed to refining our methods and expanding our analysis, with the hope that our insights will contribute to meaningful conversations and, ultimately, positive change in the world of sports and beyond.